MIL-D-17762B(SHIPS)
2 August 1965

SUPERSEDING
MIL-D-17762A(SHIPS)
22 January 1954
(See 6.3)

MILITARY SPECIFICATION

DAVITS AND WINCHES, BOAT

POWER OPERATED

1. SCOPE

1.1 Scope.- This specification covers power hoisting, gravity lowering, boat davits and winches for Naval shipboard use.

1.2 Classification.- The boat davit and winch shall be of the following types, as specified (see 6.1):

Davit

Type I - Single bank boat davit.
Type II - Nested boat davit.
Type III - Triple bank boat davit.
Type IV - Quadruple bank boat davit.

Winch

Type A - Single-divided drum winch.
Type B - Double drum winch.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
HH-L-361 - Linings, Brake (Automotive Use).
RR-W-410 - Wire Rope and Wire Strand.
TT-P-644 - Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

MILITARY

MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment For Naval Shipboard Use.
MIL-C-2174 - Controllers, Direct-Current, Naval Shipboard.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-P-3184 - Preparation For Delivery of Capstans, Winches, Windlasses, and Associated Repair Parts.
MIL-F-3541 - Fittings, Lubrication.
MIL-C-11796 - Corrosion Preventative Compound, Petrolatum, Hot Application.
MIL-P-15024 - Plates, Identification — Information and Marking For Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-E-15130 - Enamel, Ship, Exterior, Alkyd, Haze Gray, No. 27, Formula No. 5H.
MIL-B-16392 - Brakes, Magnet, Naval Shipboard.

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MILITARY (CONT.)

MIL-M-17060 - Motors, Alternating Current, Integral, Horsepower (Shipboard Use).
MIL-C-17361 - Circuit Breakers, Air, Electric, Insulated, Enclosure (Shipboard Use).
MIL-M-17413 - Motor, Direct Current, Integral, Horsepower (Shipboard Use).

STANDARD

MILITARY

MIL-STD-278 - Welding and Allied Processes.

PUBLICATIONS

BUREAU OF SHIPS

Technical Bulletin No. 5 - Instructions for Design and Care of Wire Rope Installation.

DRAWINGS

BUREAU OF SHIPS

S8201 - 848995 - Disc Bearing Hooks For Lowering Boat Falls Blocks.
S8201 - 848996 - Disc Bearing Hooks For Lowering Boat Falls Blocks.
805-1872782 - Davit, Crescent, Tensioning Device 26 Foot Motor Whaleboat (Plastic).
805-1872783 - Davit, Crescent, Tensioning Device 26 Foot Motor Whaleboat (Plastic).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN GEAR MANUFACTURERS ASSOCIATION STANDARDS

(Application for copies should be addressed to the American Gear Manufacturers Association, 302 Empire Building, Pittsburgh 22, Pennsylvania.)

3. REQUIREMENTS

3.1 Material.- Materials shall be suitable for the purpose intended. Unless otherwise specified in the contract or order materials shall be in accordance with either commercial or government specifications.

3.1.1 If commercial specifications are used the bill of material shall list both the commercial specification and the equivalent government specification.

3.1.2 Cast iron shall not be used for any part.

3.1.3 Aluminum 6061-ST shall not be used for any part.

3.2 Design.-

3.2.1 The boat davit and winch shall be complete in all respects with the exception of power leads and electric control cable between the external switches and control panel. Unless otherwise specified in the contract or order (see 6.1), the davit and winch shall be designed for deck installation. The winch shall have sufficient wire rope stowage capacity to permit lowering a boat a distance not less than 50 feet below the deck upon which installed or shall permit lowering a boat a greater distance if specified (see 6.1).

3.2.1.1 Welding and allied processes employed in the fabrication of any item or component shall be in accordance with applicable requirements of MIL-STD-278. Stress relieving shall be required for welding fabrications. Non-destructive tests of welds will not be required.

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3.2.1.2 Design of the winch and davit shall be such as to eliminate hazards of parts becoming inoperative due to frequent soaking by salt spray and extreme atmospheric conditions occurring with the winch and davit in an exposed position on the weather deck of the vessel. The winch shall be suitable for operation between ambient temperatures of 120°Fahrenheit (F) to 0°F. and a design life of not less than 1,000 hours.

3.2.1.3 The winch and davit shall be as compact and light as is consistent with adequate strength and rigidity. Maximum size and weight shall not exceed that specified (see 6.1). Full use shall be made of welded construction to obtain lightest weight.

3.2.1.4 Protective means shall be provided for securing keys, feathers, splines, nuts, screws, and collars to prevent their working loose. The use of set screws shall be avoided. When set screws are used, means shall be provided to prevent their working out.

3.2.2 Gearing.- All gearing shall be in accordance with American Gear Manufacturers Association standards and shall have machine cut teeth, shall be totally enclosed in an oiltight case, and shall be lubricated by an oil bath. The oiltight gear cases shall be provided with suitable means for filling, draining and measuring lubricant level. Enclosures for the reduction gears shall be such as to permit easy inspection and removal of internal parts.

3.2.3 Bearings.- All bearings except those lubricated by an oil bath or those which will be inaccessible for addition of lubricant after installation shall be fitted with pressure grease fittings in accordance with type I or III of MIL-F-3541. Bearings which will be inaccessible for periodic lubrication after installation shall be of a sealed type which will permit both renewal or addition of lubricant during shipyard or tender overhaul. Antifriction bearings, in accordance with FF-B-171 or FF-B-185, as applicable, shall be used throughout; except, for control and accessory shafts, bronze bushings may be used. When bushings are used, positive means shall be provided to prevent the bushing from turning and cutting off the supply of lubricant.

3.2.4 Wire rope shall be 6 x 37, fibre core, in accordance with type I, class 3, construction 6 of RR-W-410, preformed, uncoated and grease impregnated.

3.2.5 Davits.- The boat davit shall be of the gravity type with arms or heads that extend over the side of the ship for launching or retrieving a boat. These arms shall be designed so that under control of the hoisting unit they can be rolled or rotated inboard to permit stowage of one or more boats leaving the side of the ship free from any structural projections. In addition, the design shall provide for movements of these arms from the inboard position to the outboard position, with a boat, under control of the hoisting unit but without the use of ship's power.

3.2.5.1 The davit shall be capable of launching its boat (boats) of the type specified (see 6.1) with a 10 degree adverse list on the ship. Under such list the clearance between the side of the boat and the edge of the deck upon which the davit is mounted shall be not less than 6 inches.

3.2.5.2 The design of the components shall permit installation of the davit on either the port or starboard side of the ship without the substitution of parts. The boats will always be handled bow forward.

3.2.5.3 The davit shall be designed for mounting on and securing to deck pads by means of body bound bolts. Bolts and deck pads shall be furnished by the installing activity. The deck pads will be built up from the deck to compensate for both camber and shear of the deck. For design dimensioning purposes the shortest pad will not extend more than 4 inches above the surface of the deck. The davits shall be furnished complete with all structural supports for 100 percent deck mounting. However, the davit shall be equally adaptable for installations requiring attachment to a deck house to support the inboard end of the davit structure. The general arrangement drawing (see 3.10) shall include a sketch indicating the type of bracket required and method by which it should be attached to the davit structure together with a note that this bracket, if required, shall be furnished by the installing activity.

3.2.5.4 For protection against possible towing forces, the davit shall be designed to withstand a horizontal working load of a value not less than 10 percent of the hoisting weight of the boat acting either fore or aft, on the extreme outboard end of the davit arm through and normal to the longitudinal centerline of the davit arm. For type I and II davits, each arm shall be designed to sustain the entire towing load, unless semi-rigid cross bracing is provided. For type III and IV davits and those of type I and II davits having cross bracing the towing load shall be considered as distributed to both arms through the strongback or bracing.

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3.2.5.5 Watertight limit switches one for each davit arm, shall be provided on the davit structure for actuation by the davit arms to prevent powered movement of the arms inboard beyond safe limits.

3.2.5.6 The davit structure shall have inboard stops and automatic latches to secure the arms against athwartship movement when stowed in the inboard position. For type I and type II davits these stops shall be a portion of the griping arrangement. In no case shall the inboard stop be of a type that can be left in a position to permit the davit arms to be two-blocked before power has been cut off by the limit switches.

3.2.5.7 Boat falls.- The boat falls, one for each of the two boat hoisting rings, may be single or two parts purchase. Single part purchase shall not be used if the stress calculations indicate that the required wire rope size would exceed 7/8 inch diameter. Where two parts purchase are used, the lifting hook shall be attached to a single sheave floating block. Where chain links are used between the floating block and the hook they shall be closed and welded on the side.

3.2.5.8 Lifting hooks.- Lifting hooks shall be of the "Raymond Quick Release" type, in accordance with standard plan S8201-848995 and S8201-848996 (without swivel).

3.2.5.9 To reduce the tension in the boat falls to compensate for the additional load of the davit arms as the boat is being hoisted from the outboard to the inboard position, additional purchases which will come into effect for this portion of the hoisting procedure may be provided.

3.2.5.10 Shock absorbers shall be provided to minimize shock which will occur while hoisting a boat from the water in a rough sea. The shock absorbers shall allow 6 inches (minimum) of movement after the boat hoisting weight is suspended on the falls.

3.2.5.11 To permit a ready means of equalizing the boat falls, a boat fall adjustment feature shall be provided which will allow a takeup of at least 6 inches on each boat hook.

3.2.5.12 Type I and type II davits shall be furnished complete with boat gripes. Type III and type IV davits shall be furnished without boat gripes or deck stowage chocks.

3.2.5.13 All sheaves and associated housing required to lead the boat falls from the winch to, on and through the davit structure shall be furnished by the contractor. Those sheaves which will require securing to the deck shall be secured to deck pads by means of body bound bolts. Bolts and pads will be furnished by the installing activity.

3.2.5.14 All sheaves shall be mounted on ball or roller bearings and shall meet the design requirements of Technical Bulletin No. 5. Friction losses in such bearings will be considered 2 percent. All sheaves on the davit heads and floating blocks shall be provided with suitable guards for preventing the wire rope from jumping the grooves when slack line occurs.

3.3 Types of boat davits.-

3.3.1 Type I - single bank boat davit.- Type I boat davits shall be for single boat stowage. The davit arms shall be crescent in shape. The boat shall be stowed in a two-blocked position against the curved sections of the arms, and secured by means of gripes. The curved portion of the arm shall be fitted with leather covered pillow blocks against which the boat can be griped for secure stowage without damage to the boat. The boat shall rest on a keel rest when in the stowed position in order to partially relieve the load on the boat hoisting rings. Where a 26 foot motor whaleboat is stowed, the keel rest shall have a quick release feature. Full length keel rest and tripod bilge rests shall be provided for plastic boats. The center to center distance of the davit arms shall be equal to the center to center distance of the boat hoisting rings. Type I davits shall also have a falls tensioning device similar to that shown on Standard plan 805-1872782 and 805-1872783.

3.3.2 Type II - nested boat davit.- Type II boat davits shall be similar to type I boat davits except that the davit shall be capable of handling and stowing two boats simultaneously. The two boats will be of unequal length with the smaller boat having a length slightly less than the center to center distance of the lifting rings of the larger boat. The smaller boat will be stowed within the larger boat. The davit shall be designed for (a) hoisting the small boat to and holding it suspended from the davit arms while the larger boat is being hoisted from the water, (b) hoisting both boats simultaneously into stowage, (c) lowering both boats

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simultaneously from stowage to the outboard position of the davit arms, (d) holding small boat suspended from the davit arms in the outboard position while launching the larger boat and then launching the smaller boat. Under (a), (b) and (c) both boats will be unloaded. Under (d) the small boat will be empty while launching the fully loaded large boat. The davit arms shall have a center to center spacing equal to the center to center spacing of the lifting rings of the larger boat.

3.3.3 Type III - triple bank boat davit.- Type III boat davits shall be suitable for stowage of two or three boats. Two of the three boats will be stowed on the deck between the davit arms, tiered one above the other. The bottom boat will be stowed in chocks on the deck. The second boat will be stowed in deck supported, retractable chocks which bridge the bottom boat. The third boat, if required, will be stowed suspended from the boat falls, with the davit arms in the outboard position, griped in against the side of the ship at or near the elevation of the rail. The center to center spacing of the davit arms shall be such that a boat can pass between the davit arms and supporting structure with a clearance of about 6 inches at each end. A structural member, herein called a strongback, shall span the distance between the two davit arms and shall be supported at each arm on a trunnion bearing. This strongback shall be of either a boxed or open lattice construction. The boat falls for hoisting the boat shall be reeved through the strongback. The boat fall supports shall have a center to center spacing equal to the center to center spacing of the boat hoisting rings and so located on the strongback as to provide equal clearances at the bow and stern with respect to the davit arms. Means shall be provided for automatically latching the boat blocks or hooks to the strongback when hoisting the davit arms from the outboard to the inboard position and automatically unlatch while lowering the davit arms from the inboard to the outboard position. Manual release shall be provided for the boat blocks or hooks to permit lowering a boat to its deck stowage position, and at the same time retain the latching feature for automatic operation when hoisting a boat from its deck stowage position. Manual release of the davit arm latches shall be provided to permit lowering the davit arms to the outboard position.

3.3.4 Type IV - quadruple bank boat davit.- Type IV davits shall be similar to type III davits except that three boats in lieu of two will be stowed in chocks on the deck of the ship. This stowage will consist of tiered stowage of two boats, identical to the two boat stowage of the type III davit, inboard; and single boat stowage of one boat in chocks on the deck outboard of the tiered stowage. A fourth boat, if required, will be stowed suspended from the boat falls, with the davit arms in the outboard position, and the boat griped in against the side of the ship.

3.4 Boat davit winch.-

3.4.1 The boat davit winch shall be designed specifically for hoisting the boat (see table I and 6.1), to stowed position. In addition the winch shall incorporate features to permit operation of a davit of the type specified (see 6.1).

Table I - Type boat and design hoisting weight.

Type Boat	Design Hoisting Weight, Pounds.
26 Foot Motor Whaleboat	6500
26 Foot Motor Whaleboat Mk. 5 (Plastic)	6500
26 Foot Motor Whaleboat Mk. 6 (Plastic)	6500
26 Foot Personnel Boat	10,000
28 Foot Personnel Boat Mk. 6	12,000
33 Foot Utility Boat	11,000
35 Foot Motor Boat	
36 Foot Landing Craft LCVP Mk. 1	Davits for these boats shall be designed for a working load of 26,000 (min.)
36 Foot Landing Craft LCP(L) Mk. 1	
36 Foot Landing Craft LCP(L) Mk. 2	
36 Foot Landing Craft LCP(R)	

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3.4.1.1 The winch shall provide power hoisting and gravity lowering of the boat. An automatic means (centrifugal brake or similar) shall be provided to control the gravity lowering speed of the boat to a speed between 90 and 100 feet per minute. In addition a brake shall be provided which is designed for smooth controlled lowering of the boat at any speed between 20 and 100 feet per minute and to stop and hold a load equal to 200 percent of the hoisting weight of the boat. The brake lining shall be in accordance with HH-L-361.

3.4.1.2 Whenever power payout of the unloaded boat falls in a requirement, means shall be provided to prevent energizing the electric motor in the payout direction with the gravity lowering control brake set.

3.4.1.3 Hand cranks shall be provided to permit manual operation of the winch in the hoist direction under full load or in the payout direction under no load in the event of power failure. The hand crank radius shall not exceed 12 inches and the crank effort shall not exceed 60 pounds when hoisting the boat. A positive electrical interlocking device shall be provided to prevent power operation of the winch in either the hoist or payout direction with the hand cranks engaged.

3.4.1.4 Under no condition of power failure while power hoisting or during hand hoisting shall dropping of the boat be possible.

3.4.1.5 The winch shall be designed so as to protect the electric motor at all times from damage due to being driven at excessive speeds during gravity lowering of a boat.

3.4.2 Type A - single divided drum winch.- Type A boat davit winch shall be a single divided drum unit. The divided drum shall permit simultaneous control of two wire rope boat falls with one wire rope being spooled on each section of the drum. On type A winches for use with gravity type davits, each section of the drum shall be grooved and shall be of sufficient diameter and length to stow the required length of wire rope on one layer. The winch shall be capable of paying out the unloaded boat falls with the electric motor. A payout mechanism driven by the winch drum gearing shall be provided to perform the functions of the spooling device specified in 3.4.3. On type A winches for use with Crescent, boom, sheath screw and similar type davits, the drum sections shall be of the size specified (see 6.1) or of sufficient size to stow the required length of wire rope in as few layers as practical with a minimum of 2 inches of clear flange above the last layer. The winch shall be equipped with a hand spooling device to permit simultaneously controlled spooling of the wire rope on each drum section. The winch shall also be provided with a hoist limit switch which can be adjusted to prevent power hoisting beyond safe limits.

3.4.3 Type B - double drum winch.- Type B boat davit winch shall be a two drum unit. Each drum shall be mounted on a separate shaft and arranged so that wire rope leads, from the two drums to the davit, shall have a center to center separation of not more than 3 inches for any amount of wire rope on the drum. The two drums shall be mechanically locked so as to provide equal payout and recovery of wire rope on the two drums. Each drum shall be grooved and shall be of sufficient length and diameter to stow the required length of wire rope on one layer. The winch shall be capable of paying out unloaded boat falls with the electric motor. The winch shall be equipped with a power driven spooling or level winding device which shall provide the following functions (a) maintain accurate spooling of the wire rope to prevent piling up or groove jumping and (b) maintain a tension on the wire ropes between the spooling device and the winch drums to prevent the wire rope from springing out from the surface of the drum and fouling during payout.

3.5 Electrical equipment.-

3.5.1 Electric motors.- Unless otherwise specified in the contract or order (see 6.1) electric motors shall be so rated that they will not be overloaded during the vertical lift, and not overloaded more than 50 percent when bringing the boat and davit arms to stowed position. Motors and associated electrical equipment shall be either alternating current (a.c.) or direct current (d.c.) as specified.

3.5.1.1 Motors, d.c.- Motors d.c. shall be in accordance with MIL-M-17413 and the following:

Service - Navy "A"
Ambient temperature - 40° C.
Voltage - 230 volts, d.c.
Duty - Short time, 15 minutes.
Horsepower - As required
Enclosure - Watertight
Speed - As required

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Bearings - Ball
Insulation - Class B
Winding - Compound
Shunt field - Rated for continuous connection to line voltage.

3.5.1.2 Motors, a.c.- Motors a.c. shall be in accordance with MIL-M-17060 and the following:
Service - Navy "A"
Ambient temperature - 40° C.
Voltage - 440 volts, 3 phase, 60 cycle
Duty - Short time, 15 minutes
Phase characteristics - Squirrel cage induction
Horsepower - As required
Design - D
Enclosure - Watertight
Speed - As required
Bearings - Ball
Insulation - Class H
Radio noise suppression - Not required.

3.5.2 Electric motor controllers.- Electric motor controllers shall be in accordance with MIL-C-2212 for a.c. or MIL-C-2174 for d.c. as required (see 6.1) and the following:
Voltage - As required
Operation - Magnetic
Type - Across to line for a.c.
Reduced voltage for d.c.
Function - Motor starting and reversing
Performance - Non-automatic
Enclosure - Dripproof
Protection features - LVR and thermal overload with remote emergency run. Main contactors shall be provided in addition to reversing contactors.

3.5.2.1 Contactor arrangement.- Contactors shall be of sufficient number and so arranged that when the motor power supply is interrupted by the controller, the supply leads are opened in two places by contactors which are not interlocked.

3.5.2.2 Contactors for a.c. controllers.- All contactors for a.c. controllers shall be selected in accordance with table II.

Table II - Contactors for a.c. controllers.

Contactor size	Maximum motor rated horsepower at	
	220 volts, 3 phase	440 volts, 3 phase
0	1	1
1	3	5
2	10	15
3	20	30
4	30	60
5	75	150
6	150	300

3.5.2.3 Wiring of limit switches.- All controls employing davit arm inboard travel limit switches, hand crank interlock switches or brake interlock switches shall be wired as follows:

(a) The switch contacts shall be connected in series and in a common control lead.
(b) The switch contacts shall be connected on the opposite side of the contactor operating coils from the master switch contacts.

Hoist limit switches (on type A winches) shall be similarly connected in the appropriate control lead.

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3.5.3 Master switch.- Each winch shall be provided with a lever operated, spring return to "OFF" position, master switch for remote bulkhead mounting. The switch shall be enclosed in a watertight housing.

3.5.4 Emergency run.- An "Emergency Run" push button shall be provided. The emergency run feature may be in the same enclosure as the master switch or may be enclosed in a separate watertight housing.

3.5.5 Emergency disconnect switches.- Each winch shall be provided with an emergency disconnect switch to be installed adjacent to the master switch. The switch shall consist of a suitable NQB circuit breaker, in accordance with MIL-C-17361, mounted in a watertight housing with means provided for actuation of the circuit breaker from the side of the housing. The switch shall be connected in the motor supply leads between the motor control panel and the motor and, when actuated, shall function to disconnect the motor from its power supply.

3.5.6 Electric brakes.- Electric brakes, if applicable, shall be in accordance with MIL-B-16392.

3.6 Performance.- Unless otherwise specified in the contract or order, the hoisting speed while handling the boat through the vertical lift shall be not less than 40 feet per minute. The davit and winch shall be designed to both launch and retrieve the specified boat (see 6.1), from stowage to the water and from the water to stowage.

3.7 Stress calculations.- A complete set of stress calculations in duplicate for the davit and winch shall be included with the first submission of drawings. Calculations for the davits shall include the davit heads, trackways, davit head roller shafts, pins, sheaves, wire rope and trackway holding down bolts. Calculations for the winch shall include the gears, bearings, shafts, keys, holding down bolts, brakes and clutches.

3.7.1 While handling rated load through the full travel, the combined stress in any part shall not exceed 35 percent of the tensile yield point of the ductile material used nor shall the factor of safety of the wire be less than 5 based upon its minimum breaking strength.

3.7.1.1 The compressive strength of aluminum virgin metal (no weld) shall be assumed equal to the ultimate tensile strength.

3.7.1.2 The factor of safety of the wire rope shall not be less than 1.5, based upon its minimum breaking strength, when subjected to a load equal to the stalling torque of the electric motor.

3.7.2 The combined stress in no part of the winch shall exceed 70 percent of the tensile yield point when subjected to a load equal to the stalling torque of the motor.

3.7.3 In stress calculations, the following strength relationships shall be used:

Design strength	Percent of tensile yield point
Direct shear	60
Torsional shear	65
Compression (bearing)	160 (except as specified in 3.7.1.1).

3.7.3.1 Other values may be used if substantiated by tests and approved by the bureau or agency concerned.

3.7.3.2 The minimum breaking strength of the wire rope shall be the value specified in RR-W-410 and the method of calculating the factor of safety of the wire rope shall be as specified in Technical Bulletin No. 5.

3.8 Identification plates.- Identification plates and other designated marking plates shall be in accordance with type C of MIL-P-15024. This identification plate shall also contain the name of the particular article and the stock number with the words "Federal Stock Number_____". (As many as seventeen spaces

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may be required.) A separate identification plate shall be required for the davit and for the winch. Component parts which will be supplied by the manufacturer in an unassembled state shall be provided with identification plates to permit their identity as a part of the unit. The same identity of each part shall be included on the general arrangement and installation drawings.

3.9 Finish.- Prior to shipment, the winches and davits shall be thoroughly cleaned and all parts not intended to remain bright shall be given two coats of primer conforming to TT-P-644. A final coat of enamel conforming to MIL-E-15130 shall then be applied. Exposed surfaces that will be inaccessible after installation (such as undersides of bedplate) shall be given three coats of primer conforming to TT-P-644 or heavily coated with a rust preventive compound conforming to class 1A of MIL-C-11796.

3.10 Drawings.- All drawings shall be class A in accordance with MIL-D-963.

3.10.1 Preliminary drawings submitted for approval by the bureau or agency concerned shall be forwarded in accordance with the requirements of MIL-D-963. For the purpose of scheduling, the contractor may assume 21 days will be required by the approving activity on each drawing submittal.

3.10.1.1 Preliminary drawings submitted for approval shall consist of the following:

(a) General arrangement drawing.
(b) Assembly and sub-assembly drawings.
(c) Electric motor, controller and brake drawings.
(d) Stress calculations (complete set).
(e) Installation drawings.
(f) Center of gravity indicated and dimensioned.
(g) Weight notation.

3.10.2 General arrangement drawing shall contain the following in addition to the assembly views:

(a) A list of onboard repair parts furnished.
(b) An index of related drawings with the contractors number and title.
(c) The line speed and pull at winch drums (on winch general arrangement).
(d) The maximum load on the wire rope between the davits and the winch (on the winch general arrangement).
(e) The length of wire rope which can be stowed on the drums (on winch drawing).
(f) Working load for which the davits were designed (on davit drawing).
(g) The center of gravity shall be indicated and dimensioned (davits and winch; no boats).

3.10.3 The installation drawing shall show the installation arrangement for the davit, winch, and deck sheaves, if applicable. This drawing shall contain complete information required for shipboard installation and shall include data required to permit design of foundations and deck pads. The maximum rated loading and its direction of force shall be indicated for each foundation and deck pad.

3.10.4 Final drawings shall consist of all assembly, subassembly, and detail drawings.

3.10.5 Any work performed by the contractor, and any material acquired in advance of the approval by the Government of the preliminary drawings and stress calculations shall be at the contractor's risk.

3.10.6 Neither Government approval of drawings nor waiver of such action shall relieve the contractor of his obligation to meet requirements specified in the contract and the applicable equipment specification.

3.11 Manuals.- Manuals shall conform to type I of MIL-M-15071. Two preliminary copies of the manual shall be forwarded for approval and assignment of the bureau or agency concerned identification numbers. These preliminary manuals shall contain a print of all the drawings listed below:

(a) General arrangement drawing and installation drawing.
(b) Cross-sectional assembly and sub-assembly drawings in sufficient number to adequately identify parts.
(c) Bedplate or foundation drawing.
(d) Lubrication charts with lubricants listed according to Government specification number or symbol number.

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(e) Operating instructions.
(f) Complete maintenance instructions including gear tooth clearance, and so forth.
(g) Electric motor and controller drawings and certification data.

3.12 Onboard repair parts.-

3.12.1 Mechanical.- Unless otherwise specified in the contract or order (see 6.1), each set of mechanical repair parts shall consist of the following items, as applicable, in sufficient quantity for one davit and winch:

(a) One set of bushing type bearings, if applicable.
(b) One set of oil seals and special packing.
(c) One set of antifriction bearings.
(d) One set of brake lining and rivets.
(e) One set of clutch lining and rivets, if applicable.
(f) One complete set of springs.
(g) One set of boat falls (wire rope and end fittings).
(h) One floating block of each size and type.
(i) One set of bevel gears, (if used in the winch design).

3.12.2 Electrical.- Electrical repair parts shall be furnished in accordance with MIL-C-2174, MIL-C-2212, MIL-B-16392, MIL-M-17060 and MIL-M-17413, as applicable, except that no special tools shall be provided.

3.13 Special tools.- Special tools and wrenches necessary for the proper service maintenance of the equipment shall be furnished. Each tool and wrench shall be indelibly marked for the service for which it is intended.

3.14 Workmanship.- The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.-

4.2.1 Lot.- All units of the same type offered for delivery at one time shall be considered a lot. In the following text the term "set" is intended to mean one complete installation including all associated fittings for handling the boat or boats.

4.2.2 Sampling for tests.- From each lot completely assembled davits and winches shall be selected in accordance with table III. The sample davits and winches shall be examined as specified in 4.3 and subjected to the tests specified in 4.4 through 4.4.4.

4.3 Visual examination.- Each sample set of boat davits shall be examined to verify conformance to the workmanship and other requirements of this specification which do not involve tests. Each sample assembled winch shall be similarly examined. If any defect is found in any sample equipment, every equipment in the lot shall be examined for the same defect. All defective samples shall be rejected.

4.4 Tests.- For the tests specified in 4.4.2 and 4.4.3 the load shall be lowered and raised through a vertical distance of at least 20 feet.

4.4.1 Operation of winch without load.- Each winch shall be run under no load for a period of one-half hour in each direction, if applicable. No abnormal heating of any part shall occur. The spooling and tensioning device may be disconnected for this test.

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Table III - Sampling for examination and tests.

Number of davits or winches in lot	Number of davits or winches in sample
1 through 4	1
5 through 12	2
13 through 25	3
26 through 49	4
50 through 99	5
100 and over	6

4.4.2 List test.- One set of davits and winch of each rating and type shall demonstrate satisfactory operation when set up to simulate a 10 degree list outboard and a 10 degree list inboard.

4.4.3 Load test.- Each sample set of davits and winch shall hoist and lower rated load at rated speed and 150 percent load at no specified speed through a complete cycle to and from stowed position.

4.4.4 Static test.- Each sample shall be subjected to a static load equal to 200 percent rated load with the davit arms in the outboard position for a period of 10 minutes. No permanent set shall result. This load shall then be lowered and intermittently stopped under control of the gravity lowering brake. No permanent set shall result.

4.5 Shipboard test.- After installation aboard ship, each unit shall be tested as specified in 4.4.1 and then the winch and davit subjected to the tests specified in 4.4.2, 4.4.3 and 4.4.4 to determine conformance with this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.-

5.1.1 The equipment, accessories, technical publications and repair parts shall be preserved and packaged by level A or C; packed by level A, B or C and marked levels A, B or C as specified (see 6.1) in accordance with MIL-P-3184.

6. NOTES

6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of winch and davit required (see 1.2).
(c) Whether davit and winch are to be designed for other than deck installation (see 3.2.1).
(d) Vertical distance from the water line to the top of the winch k foundations with the ship on an even keel (see
(e) Size and weight of unit (see
(f) Description of boat (boats to be handled) (see 3.4):
  (1) Type of boat.
  (2) Center to center distance of hoisting rings.
  (3) Overall length.
  (4) Beam.
  (5) Hoisting weight.
  (6) Depth from top of gunwale to bottom of keel amidships.
(g) Size of drum required.
(h) Electric motor, and motor controller, ordering data as specified in the applicable electrical specification (see 3.5)
(i) Repair parts, if other than specified (see 3.12.1).
(j) Bid data when required (see 6.2).
(k) Selection of applicable level of preservation, packaging, packing and marking required (see 5.1).

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6.2 Bid data.- Bid data, when required (see 6.1) should include the following:

(a) An approximate dimensioned general arrangement drawing and diagrammatic drawing of the davit and the winch showing all gearing and component parts.
(b) A complete description of operation of the davit and the winch with particular attention to the operational features specified herein.
(c) Estimated weight of complete davit and winch.
(d) Manufacturer's identification of proposed electric motor control panel, and all external switches.

6.3 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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